ALFIDO TECH INTERNSHIP

Task 1 — Temperature Monitoring & LCD Display

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Task	Task 1
Platform	Wokwi Arduino Simulator
Controller	Arduino UNO
Temperature Sensor	DS18B20
Display	16×2 I2C LCD
Simulation Output	22.00 °C
Project Status	Successfully simulated and tested
Submission Date	1 September 2026

This submission documents a browser-based simulation of a temperature monitoring system. The project reads temperature from a DS18B20 sensor using an Arduino UNO and displays the measured value on a 16×2 I2C LCD. The simulation was used because physical hardware components were not available.

1. Objective

To design and simulate a basic embedded temperature monitoring system using an Arduino UNO, DS18B20 digital temperature sensor and I2C LCD. The system continuously reads the temperature, shows it on the LCD and also sends the reading to the Serial Monitor.

2. Components / Technologies Used

Component / Technology	Purpose
Arduino UNO	Microcontroller and main control unit
DS18B20	Digital temperature sensing
4.7 kΩ resistor	Pull-up resistor for DS18B20 data line
16×2 I2C LCD	Displays measured temperature
C/C++ / Arduino	Embedded programming
Wokwi	Online circuit simulation and testing
OneWire + DallasTemperature	DS18B20 communication
LiquidCrystal_I2C	I2C LCD control

3. Circuit Connections

Device Pin	Arduino UNO
DS18B20 VCC	5V
DS18B20 GND	GND
DS18B20 DQ / Data	D2
4.7 kΩ resistor	DQ/Data ↔ 5V
LCD GND	GND
LCD VCC	5V
LCD SDA	A4 (SDA)
LCD SCL	A5 (SCL)

The DS18B20 uses a 1-Wire data connection on D2. The LCD uses A4 (SDA) and A5 (SCL). The I2C LCD is configured at address 0x27.

4. Circuit Simulation Screenshot

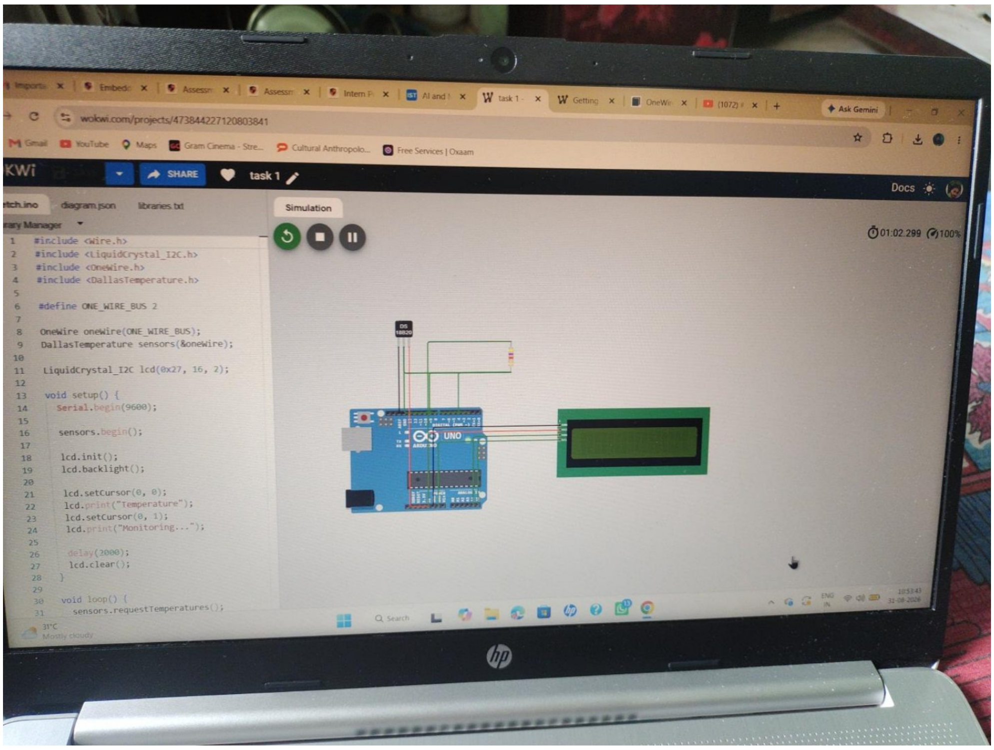

Figure 1 — Simulated circuit and component wiring.

5. Arduino Source Code

Source code used in the submitted Wokwi project:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <OneWire.h>
#include <DallasTemperature.h>

#define ONE_WIRE_BUS 2

OneWire oneWire(ONE_WIRE_BUS);
DallasTemperature sensors(&oneWire);

LiquidCrystal_I2C lcd(0x27, 16, 2);

void setup() {
Serial.begin(9600);
sensors.begin();
lcd.init();
lcd.backlight();
lcd.setCursor(0, 0);
lcd.print("Temperature");
lcd.setCursor(0, 1);
lcd.print("Monitoring...");
delay(2000);
lcd.clear();
}

void loop() {
sensors.requestTemperatures();
float temperature = sensors.getTempCByIndex(0);

lcd.setCursor(0, 0);
lcd.print("Temperature:");
lcd.setCursor(0, 1);
lcd.print(temperature, 2);
lcd.print((char)223);
lcd.print("C ");

Serial.print("Temperature: ");
Serial.print(temperature);
Serial.println(" C");

delay(1000);
}
```

6. Working Principle

1. Arduino initializes the DS18B20 sensor and I2C LCD.

2. The DS18B20 measures the current temperature.

3. Arduino reads the temperature in degrees Celsius.

4. The value is displayed on the 16×2 LCD.

5. The same value is printed to the Serial Monitor.

6. The process repeats approximately once every second.

7. Successful Simulation Output

The simulation was successfully executed. The LCD displayed 22.00 °C and the Serial Monitor reported the temperature.

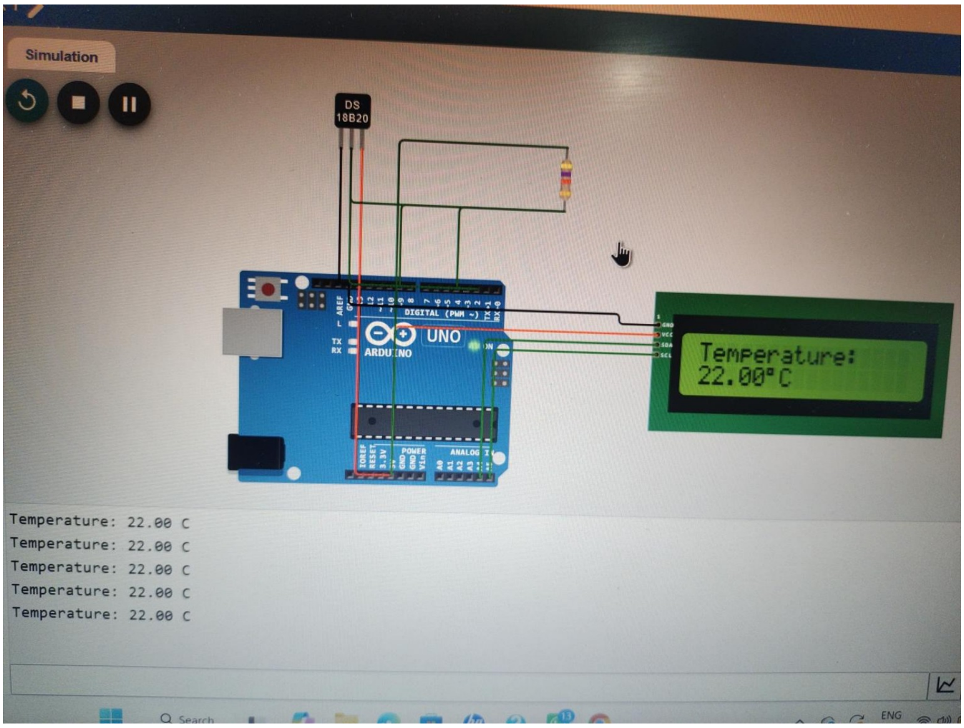

Figure 2 — Successful simulation showing 22.00 °C on LCD and Serial Monitor.

8. Result

The temperature monitoring system was successfully designed, simulated and tested. The DS18B20 reading was obtained by the Arduino UNO and displayed correctly on the I2C LCD. The Serial Monitor also confirmed the temperature reading.

9. Wokwi Project / Demo Link

https://wokwi.com/projects/473844227120803841

10. Evaluation Checklist

Requirement	Status
Code compiles and runs	✓ Completed
Proper wiring	✓ Verified in simulation
README / documentation	✓ Included
Wiring diagram	✓ Included
Short demo / output proof	✓ Simulation screenshot included
Public project link	✓ Included

Declaration: This Task 1 was implemented and tested as a virtual simulation in Wokwi because physical hardware components were not available.